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A) Normal Circuit (series)

$$1) 20 + 30 + 50 = 100 \Omega$$

$$2) I = \frac{V}{R} = \frac{125}{100} = 1.25 \text{ A}$$

$$3) 1.25 \text{ A} \quad I_1 = I_2 = I_3$$

$$4) V = IR \rightarrow 1.25(20 + 30 + 50) = 125 \text{ V}$$

$$V_{20\Omega} = 1.25(20) = 25 \text{ V}$$

$$5) \cancel{R \rightarrow V} \quad V_{30\Omega} = 1.25(30) = 37.5 \text{ V}$$

$$V_{50\Omega} = 1.25(50) = 62.5 \text{ V}$$

$$5) P = VI \quad \cancel{25 \times 1.25} \quad 1.25(25 + 37.5 + 62.5) = 156.25 \text{ W}$$

$$V_{20\Omega} = 1.25 \times 25 = 31.25 \text{ W}$$

$$V_{30\Omega} = 1.25 \times 37.5 = 46.875 \text{ W}$$

$$V_{50\Omega} = 1.25 \times 62.5 = 78.125 \text{ W}$$

B) Parallel circuit

$$1 \quad \frac{1}{20} + \frac{1}{100} + \frac{1}{50} = \frac{8}{100}$$

$$R_{eq} = \frac{1}{\left(\frac{8}{100}\right)} = 12.5 \Omega$$

$$2 \quad I = \frac{V}{R} \quad \frac{125}{20} + \frac{125}{100} + \frac{125}{50} = 10 \text{ A}$$

$$3 \quad I_{20\Omega} = \frac{125}{20} = 6.25 \text{ A}$$

$$I_{100\Omega} = \frac{125}{100} = 1.25 \text{ A}$$

$$I_{50\Omega} = \frac{125}{50} = 2.5 \text{ A}$$

$$4 \quad V_{20} = V_{100} = V_{50\Omega} = 125 \text{ V}$$

$$5) P = VI \quad 125(6.25) = 781.25 \text{ W (P}_{20\Omega})$$

$$125(1.25) = 156.25 \text{ W (P}_{100\Omega})$$

$$125(2.5) = 312.5 \text{ W (P}_{50\Omega})$$